Customer Segmentation from Transactions

RFM-Based Analysis Using Unsupervised Machine Learning

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1 Introduction and Problem Formulation

Retailers accumulate transaction histories that contain information about how recently, how frequently and how much customers purchase. Analysing every transaction independently does not directly provide a concise view of customer-level behaviour. Customer segmentation addresses this by grouping customers with similar characteristics, potentially supporting differentiated engagement strategies.

The objective of this project is to identify purchasing-behaviour groups in an online retail dataset using unsupervised machine learning. Each customer is represented by three RFM variables: Recency (days since the latest purchase), Frequency (number of distinct invoices) and Monetary (total retained purchase value). Since the dataset contains no ground-truth segment labels, the task is clustering rather than classification.

Two internal validation metrics are used on the held-out customers. The silhouette coefficient compares within-cluster cohesion with separation from the nearest alternative cluster; larger values are preferred. The Davies–Bouldin index compares within-cluster dispersion with separation between clusters; smaller values are preferred. Assignment stability is separately examined with the Adjusted Rand Index (ARI), which measures agreement between two partitions after correcting for chance. These metrics assess geometric structure and repeatability, not commercial impact.

2 Dataset and Preprocessing

2.1 Source and cleaning

The study uses the UCI Online Retail dataset (Dataset ID 352), containing 541,909 invoice-line records from 1 December 2010 to 9 December 2011. The source fields include invoice number, stock code, description, quantity, invoice date, unit price, customer ID and country. UCI identifies the dataset as CC BY 4.0; the source data are not redistributed in this project repository [1].

Rows without a customer ID were removed, as were cancelled invoices (invoice numbers beginning with C), non-positive unit prices, non-positive quantities and exact duplicate rows. Missing product descriptions were not by themselves treated as grounds for deletion. The sequential cleaning counts are shown in Table 1.

Table 1: Transaction counts during data cleaning.

Cleaning stage	**Rows remaining**
Original dataset	541,909
Remove missing CustomerID	406,829
Remove cancelled invoices	397,924
Remove non-positive UnitPrice	397,884
Remove non-positive Quantity	397,884
Remove exact duplicates	**392,692**

The cleaned data contain 4,338 unique customers. Because negative quantities and cancelled invoices were excluded, Monetary represents retained positive-purchase value, not net spend after returns or refunds.

2.2 RFM feature construction

For customer i, the three features are defined as

$$R_i = d_{\text{ref}} - \max_j(d_{ij}), \tag{1}$$

$$F_i = |\{\text{distinct invoice numbers for customer } i\}|, \tag{2}$$

$$M_i = \sum_j q_{ij} p_{ij}, \tag{3}$$

where d_{ref} is one day after the latest retained transaction date, d_{ij} is a transaction date, and q_{ij} and p_{ij} are quantity and unit price. Monetary values are in pounds sterling. Frequency counts distinct invoices, not transaction rows.

The raw RFM distributions are strongly right-skewed, particularly Frequency and Monetary. Each feature was transformed using $x' = \log(1 + x)$. The customer records were split into training and test subsets in an 80:20 ratio with random state 42. This produced 3,470 training customers and 868 test customers. A StandardScaler was fitted only on training data and then applied to both subsets, avoiding the use of held-out observations when estimating scaling parameters. There are no labelled classes, so conventional class balance is not defined.

3 Methodology

3.1 Clustering models

K-Means was used as the baseline. It partitions observations into K clusters by minimising the within-cluster sum of squared Euclidean distances:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2, \tag{4}$$

where C_k is cluster k and μ_k is its centroid. Values of K from 2 to 10 were investigated using inertia and silhouette score. The selected baseline was $K = 2$.

A Gaussian Mixture Model was evaluated as an alternative. It represents the feature distribution as a weighted sum of Gaussian components:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k), \qquad \sum_k \pi_k = 1. \tag{5}$$

The number of components (2–10) and covariance structures (full, tied, diagonal and spherical) were compared using the Bayesian Information Criterion (BIC). The selected candidate had 10 components and full covariance. This BIC selection is within the GMM candidate set; it is not a direct comparison with the K-Means objective.

3.2 Experimental design

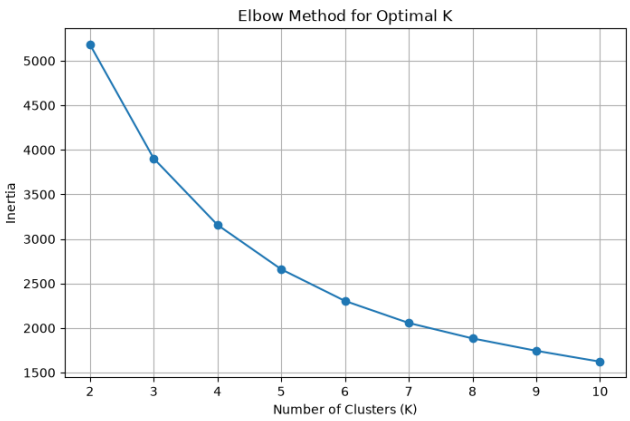
Both selected models used the same training/test partition and the same log-transformed, training-standardised RFM representation. Test-set silhouette and Davies–Bouldin scores were calculated from the test assignments. Stability was assessed separately by fitting each algorithm over 10 random initialisations on the training data and calculating pairwise ARI across the resulting partitions. This is a run-to-run stability experiment, not 10-fold cross-validation or ten independent test evaluations.

The principal improvement over clustering raw transaction rows is customer-level RFM feature engineering: transactions are first summarised into interpretable customer behaviour variables. Log transformation and training-only standardisation support the modelling process but do not add new customer information.

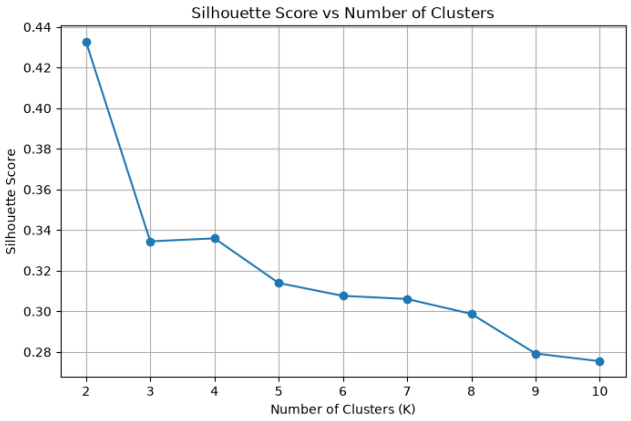
4 Results and Discussion

4.1 Selecting the K-Means cluster count

Figure 1 shows the recorded inertia and silhouette trends for $K = 2$ through $K = 10$. Inertia decreases as more clusters are added, so it must be interpreted alongside a measure such as silhouette. The highest recorded silhouette occurs at $K = 2$ (approximately 0.433), supporting the selected two-cluster baseline; the elbow curve alone does not show an unambiguous sharp elbow.



(a) Inertia across candidate K.



(b) Silhouette score across candidate K.

Figure 1: K-Means model selection on the training feature representation.

4.2 Held-out model comparison

Table 2 summarises the held-out results. K-Means obtains a silhouette score of 0.4331 compared with 0.0842 for GMM. Its Davies–Bouldin index is also lower (0.8937 versus 2.3906). Both metrics therefore favour K-Means for this dataset representation and evaluation setup. The result does not prove that K-Means is universally superior; it indicates that the tested GMM configuration was less effective in this experiment.

Table 2: Held-out clustering metrics (higher silhouette and lower Davies–Bouldin are preferred).

Model	**Configuration**	**Silhouette ↑**	**Davies–Bouldin ↓**
K-Means	$K = 2$	**0.4331**	**0.8937**
GMM	10, full covariance	0.0842	2.3906

4.3 Stability analysis

Across the 10 tested random initialisations, K-Means achieved mean pairwise ARI 1.0000 with standard deviation 0.0000. GMM achieved mean pairwise ARI 0.8941 with standard deviation 0.0571. Thus, K-Means returned identical training assignments across these runs, whereas GMM assignments varied to some extent. This is evidence of initialisation stability under the tested settings, not evidence of generalisation to new time periods or populations.

4.4 Customer segment interpretation

The selected K-Means model produced the profiles in Table 3. Cluster 0 contains 1,671 customers (38.5%) with mean recency 26.05 days, mean frequency 8.43 invoices and mean monetary value of £4,529.87. Cluster 1 contains 2,667 customers (61.5%) with mean recency 134.19 days, mean frequency 1.67 invoices and mean monetary value of £494.11.

Table 3: Mean RFM profiles of the K-Means clusters.

Cluster	**Customers**	**Recency (days)**	**Frequency**	**Monetary (£)**
0	1,671	26.05	8.43	4,529.87
1	2,667	134.19	1.67	494.11

Cluster 0 can be described as recently active, higher-value customers, while Cluster 1 represents less recent, lower-frequency customers. The labels are descriptive interpretations of cluster averages, not predefined customer classes. A PCA projection of the three standardised RFM variables is shown in Figure 2. The first two components explain 75.11% and 18.74% of variance respectively (93.85% combined). The plot gives a two-dimensional view of the separation, while the quantitative metrics remain the primary comparison.

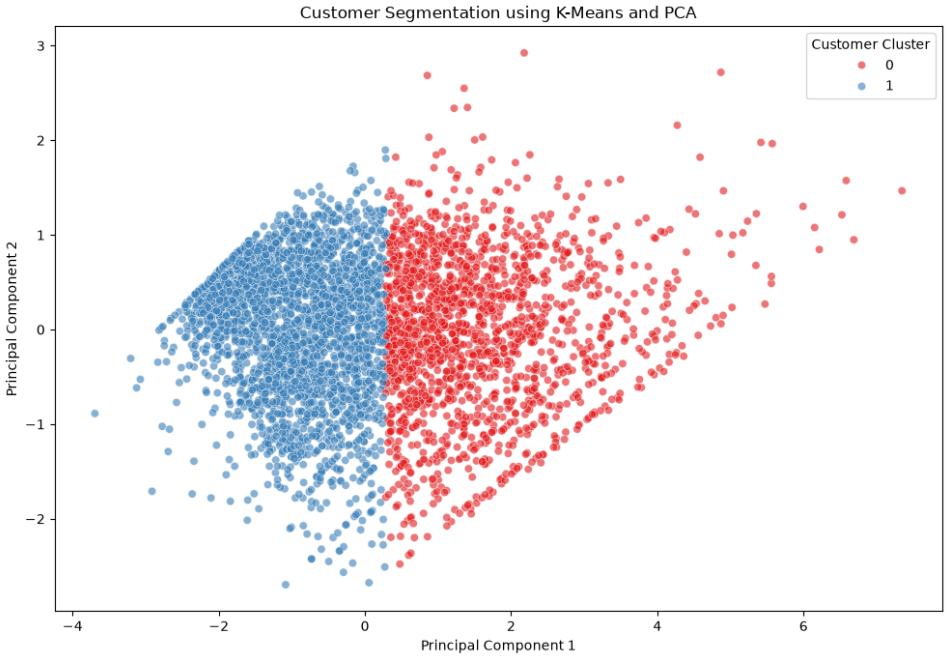


Figure 2: K-Means customer assignments projected onto the first two principal components.

4.5 Error analysis and business interpretation

The weak GMM silhouette (0.0842) suggests that its test assignments have limited cohesion/separation under the chosen representation. Its higher Davies–Bouldin index supports the same interpretation. Possible explanations include a mismatch between the selected Gaussian mixture and the geometry of the transformed RFM data, or excessive complexity from using 10 full-covariance components. These are plausible explanations, not causes established by the current experiment.

There are no ground-truth segment labels, so customers cannot be described as misclassified. A dedicated inspection of low-confidence GMM posterior assignments and K-Means customers close to the decision boundary was not performed. Consequently, individual failure examples cannot be reported honestly from the available measurements. As potential business hypotheses, Cluster 0 could be considered for loyalty and retention programmes, while Cluster 1 could be considered for carefully targeted re-engagement. These actions require separate campaign testing; the clustering results alone do not establish business impact.

5 Limitations, Ethics and Conclusion

The dataset represents one retailer over a historical period (2010–2011), limiting generalisation to current customers or other businesses. Excluding cancelled invoices and negative quantities means the Monetary feature does not account for returns/refunds. The analysis uses only three RFM variables and may overlook other factors affecting purchasing behaviour. Internal metrics assess geometric structure, not whether segments increase revenue.

Customer clusters should not be treated as personal identities or as reliable indicators of sensitive characteristics. They should not be used as the sole basis for decisions that unfairly restrict customer access or offers. Any real deployment should follow applicable privacy requirements and monitor unintended bias. From a methodological perspective, the scaler was fitted only on training data; future feature selection or dimensionality reduction used for evaluation must also be fitted without using held-out test observations.

In conclusion, customer-level RFM features provided a compact representation for exploratory segmentation of the Online Retail customers. Among the evaluated configurations, K-Means with two clusters performed better than the selected GMM on both held-out internal metrics and was more stable across the tested initialisations. The two resulting groups differ substantially in recency, invoice frequency and retained purchase value. Further work should examine temporal validation, alternative feature sets and cluster-boundary uncertainty, and experimentally test whether segment-specific campaigns provide measurable benefits.

References

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